



Predictive Model of ICU Length-of-Stay Following Surgery for Soft Tissue Infection in Sepsis

Heather Qiu, BS¹, Scott M. Ye, BS², Rachel Grant, BS³, Jacqueline Jones, MD⁴, Hyunjung Kwak, PhD^{5,6,*}, Zachary Ginsberg, MD, MPP^{7*}

¹Duke University Masters in Interdisciplinary Data Science, Durham, NC

²University of North Carolina-Chapel Hill, Department of Biostatistics, Chapel Hill, NC

³Campbell University School of Osteopathic Medicine, Buies Creek, NC

⁴Duke University School of Medicine, Department of Pediatrics, Durham, NC

⁵Massachusetts General Hospital, Boston, MA; ⁶Harvard Medical School, Boston, MA

⁷Duke University School of Medicine, Department of Emergency Medicine and Critical Care, Durham NC

*Co-corresponding and co-senior authors: Hyunjung Kwak at hkwak1@mgh.harvard.edu and Zachary Ginsberg at zachary.ginsberg@duke.edu

Background

Septic patients with skin and soft tissue infections are high resource utilizers, particularly when requiring procedures and ICU care. The length of stay in the ICU is variable, and no tool exists for this population to use initial morbidity to predict ICU length-of-stay (LOS). With pressure to reduce ICU bed scarcity in the United States, such a LOS predictive model can optimize bed flow by accounting for individual patient factors.

Methods

Using the MIMIC-IV database¹ representing adult critical care patients at a tertiary medical center we identified 439 patients between 2008-2019 who met Sepsis-3² criteria and underwent a specified procedure while in the ICU. The procedures were identified based on specific ICD-9 codes related to “amputation”, or “debridement”, or permutations of “incision and drainage”, or “necrotizing fasciitis”.

Building upon existing prediction models (NSQIP³ and LRINEC⁴), we collected physiologic variables upon ICU admission and final values at ICU discharge (pH, lactate, base excess, creatinine, ventilator status, FiO₂ status, vasopressor use, sodium, glucose, hemoglobin, white blood cell count (WBC), and C-reactive protein). Variables collected were the most severe values within 24 hours of admission to the ICU, with the exception of ventilator and FiO₂ status, of which sampling within the first 6 hours of ICU admission was used. We then used clustering method to confirm clinician gestalt of drivers of LOS and multivariate linear regression analysis to create a prediction model.

Our primary outcome was ICU LOS and secondary outcomes were survival, ICU readmission, time from ICU admission to first surgery, time from first surgery to ICU discharge, and time from first to final surgery in the ICU. We additionally defined “floor readiness” criteria to ascertain if meeting these criteria prior to floor transfer was predictive of reduced ICU readmission within 30 days.

References

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Results

Table 1: Demographics	Overall*	Low-risk for prolonged ICU stay	High-risk for prolonged ICU stay	Non Survivors of 1st ICU admission
Age (years)	61 [48, 71]	62 [52, 72]	57 [42, 70]	69 [59, 79]
Sex (# and %)				
Male	279 (65%)	140 (61%)	139 (69%)	21 (68%)
Female	160 (35%)	72 (39%)	81 (31%)	10 (43%)
Unidentified	0 (0%)	0 (0%)	0 (0%)	0 (%)
ICU Duration (days)	5 [2, 11]	3 [2, 6]	7 [4, 14]	8 [4, 18]
Time from ICU admission to first surgery (days)	0 [0, 1]	0 [0, 1]	0 [0, 1]	0 [0, 3.5]
Time from first surgery to ICU discharge (days)	0	2 [1, 4]	5 [2, 10]	6 [3, 12]
Meeting “floor readiness” criteria (%)**	22%	38%	8%	6.5%
ICU Readmission (#, %)	13 (3%)	10 (4%)	3 (1%)	n/a
Overall survival (%)	59%	54%	73%	0%

Table 1. Demographics. Continuous variables are represented by mean and IQR [25%, 75%]. For all groups, the median number of surgeries during the hospitalization was 1 [1, 1], and as such there were no days between first and final surgery while in the ICU (median 0 days [0, 0]).

*Values for “Overall” excludes patients who did not survive the first ICU admission.

**“Floor Readiness” Criteria was defined as meeting all values: pH >7.3, Lactate or lactic acid < 3 g/dL, base excess > 3 mmol/L, Creatinine <1.6 mg/dL or <1.5 x baseline, off the ventilator and FiO₂ < or = 60%, off vasopressors, sodium 135-145 mmol/L, Hgb >7 g/dL, WBC 4.5-12 x 10⁹/L, CRP < 2 mg/dL, and glucose 60-180 mg/dL.

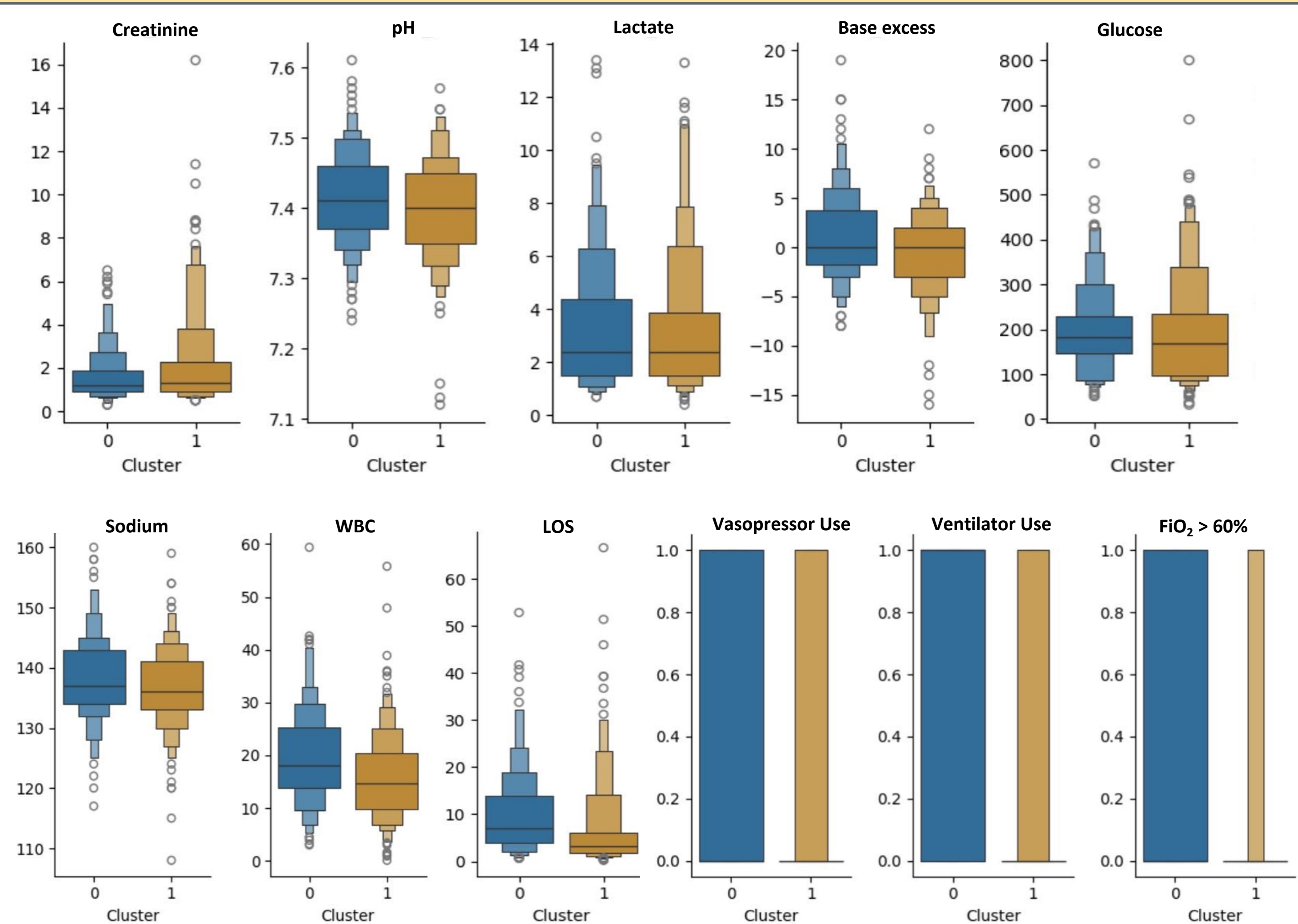


Figure 1. Boxen plots. Cluster 0 (blue) represents the high-risk group whereas cluster 1 (yellow) represents the low-risk group. Box width represents number of patients, and vertical axis demonstrates range of distribution. Binary data is represented on a scale of 0-1. Width is indicative of a larger patient concentration.

Top (left to right): Initial creatinine, Initial pH, Initial lactate, Initial base excess, Initial glucose.

Bottom (left to right): Initial sodium, Initial WBC, LOS, Initial vasopressor use, Initial ventilator use, Initial FIO₂ above 60%.

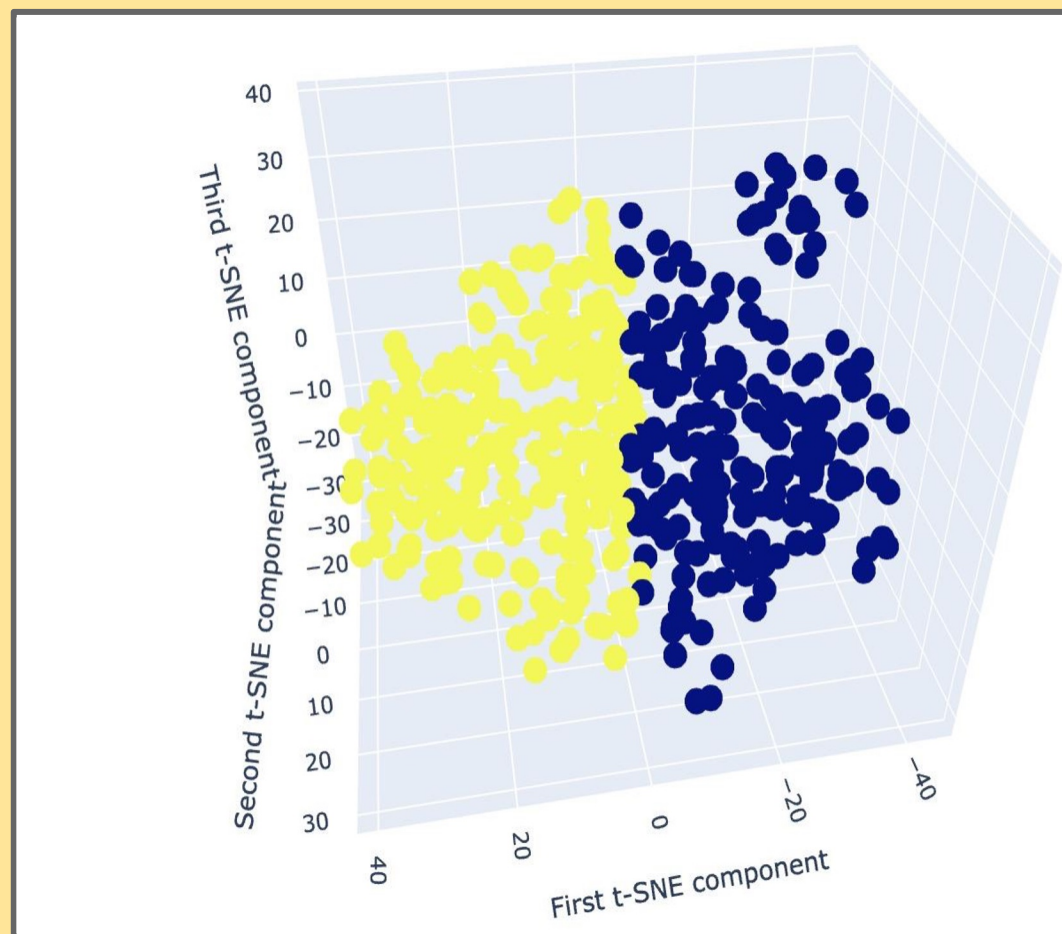


Figure 2. Cluster diagram. This demonstrates that cluster 1 (yellow) representing the low-risk group appears to be consistently better based on the tested components.

Clustering components included LOS, vasopressor use, ventilator use, FiO₂ above 60%, and initial and final sodium, WBC, creatinine, pH, lactate, base excess, glucose, CRP, as well as last hemoglobin.

Discussion

These results demonstrate preliminary finding based on efforts from a focused weekend designing the clinical question and obtaining initial results. Further work is needed to complete the predictive model, such as mediation analysis and causal inference. Our training set had a 70% accuracy on the test data.

Initial results however do provide insight in describing the current state of critically ill septic patients requiring procedural intervention for soft tissue and skin infections. Our results indicated an overall male predominance in all groups and distribution of male and female percentages were relatively consistent between the low- and high- risk groups. Our findings also demonstrate an overall median ICU duration of 5 days, assuming survival for the first ICU admission, with the low-risk group having a median of 3 days and high-risk group of 7 days (Table 1). Over 95% of patients survived the first ICU admission regardless of ICU duration.

Surprisingly, despite the low-risk group meeting more “floor readiness” criteria and having a shorter ICU stay, the low-risk group had higher ICU readmission rates and lower overall survival. We question whether the variables used in the “floor readiness” criteria may actually inappropriately provide false reassurance to the clinician in determining overall patient trajectory. The top 3 reasons for failing “floor readiness” criteria, starting with most common, was WBC count, base excess, and sodium levels.

In reviewing survival, patients who died during the first ICU admission had a larger IQR in timing of first procedure (though median remained 0 days), and a relatively prolonged ICU course following the first surgery (6 days), suggesting that rather than immediate mortality, this subgroup suffered a complex, prolonged, and ultimately unsurvivable course.

Conclusions

In patients presenting with critical sepsis requiring procedural intervention for skin and soft tissue infections, median ICU LOS for survivors is 5 days with interquartile ranges of 2-11 days. Further analysis is needed to create a predictive model using readily available clinical variables to predict ICU LOS on an individual level that can help optimize bedflow, while simultaneously providing reassurance to the clinician of a low risk for ICU readmission or other post-ICU decompensation on the floor.

Future steps: Proceeding with a causal inference statistical analysis to identify physiologic variables associated with ICU LOS and improve predictive accuracy (currently 40%) of “readiness for floor” criteria to leave the ICU without a major decompensation event within 30 days.

Limitations: This dataset was based on retrospective data from patients from a single center (Beth Israel Deaconess Medical Center in Boston, MA) and as such, application of our results may differ in hospitals with different patient demographics or hospital infrastructure. Data was collected prior to the COVID-19 pandemic and it is unknown how the post-COVID-19 era affects our findings. We defined our cohort using ICD-9 procedure codes which is intrinsically subject to error based on accuracy of clinician billing and documentation.